

Balances inside the covenant

An OP_PUSH_TX covenant that holds
the reserve and the ledger — and can't
be drained



In-covenant

THE LEDGER IS THE UTXO'S
STATE



Signature

PROVES WHO OWNS WHAT



OP_PUSH_TX

THE SPEND CONSTRAINS ITSELF



No forgery

NO TOKEN EXISTS TO FORGE

The pool is one covenant UTXO: its satoshi balance is the reserve, and its locking script carries a ledger of every holder's balance. BSV has no global mutable state, so the covenant carries its own — and forces each spend to re-lock the successor pool with the state correctly updated. This is the trick that lets a script behave like a stateful contract on a pure-UTXO chain.

Balances live **inside** the covenant as a HashMap (owner → amount), so there is no separate, forgeable token object. A buy credits an entry; a sell requires the holder's signature and debits it. Because nothing spendable leaves the pool, a forged token can't be sold back to drain the reserve — the safety property that makes this trustless with **no operator key and no indexer**.

— What the covenant enforces



Self-replication

- The spender pushes this transaction's own preimage.
- Tx.checkPreimage proves it — so the covenant reads its own tx.



The ledger is state

- A HashMap of owner-hash → balance, committed in the script.
- Buys set +delta; sells set -amount; a Merkle proof per touch.



The curve is math

- Cost and refund computed in Script, exact integer arithmetic.
- Every rounding falls in the pool's favour — never drainable.



Sells need a signature

- checkSig against the holder's key authorises the debit.
- You can only ever move your own balance.

— sCrypt → on-chain effect

SCRIPT PIECE	WHAT IT DOES ON-CHAIN
Tx.checkPreimage	Proves the pushed bytes are THIS transaction
@state HashMap	Holder balances committed inside the pool script
hash256(outputs)	Forces the spend to re-lock the successor pool
ledger.set / canGet	Update one balance with a Merkle access proof
checkSig(sig, pub)	Only the balance owner can debit it (sell)
$k \cdot \Delta \cdot (2s + \Delta + 1) / 2$	Exact curve cost — the /2 never truncates
ANYONECANPAY	Lets the trader attach their own funding input

No separate token means nothing to forge. The reserve is drain-proof by construction — enforced entirely on-chain, with no platform key and no trusted indexer.

Thank you

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